Pimpri-Chinchwad Education Trust's
Pimpri-Chinchwad College of Engineering
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)
ISO 21001:2018 Certified by TUV

Assignment No 8

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Subject: Design and Analysis of Algorithms
Batch: A3
GitHub Link: https://github.com/SOHAM0007-CODER/DAA_ASSIGNMNET

1. Topic

Find a SwiftShip delivery tour that starts at a chosen city, visits every other city once, and returns to the start. Minimize fuel cost within a total travel-time deadline using Least-Cost Branch and Bound instead of checking every city order without pruning.

2. Course Objective

- Understand the computational complexity of algorithms.
- Select a suitable algorithm design strategy for a practical problem.

3. Course Outcome

- Apply admissible bounds and a min-priority queue to exact TSP search.
- Recover the optimal feasible tour and analyze pruning and worst-case cost.

Theory

The Traveling Salesperson Problem seeks a tour through all cities with one final return to the start. Each road has a distance, fuel-consumption rate and travel time. The program minimizes total fuel expense, while the complete tour, including its return edge, must meet the time limit.

Least-Cost Branch and Bound stores partial routes in a min-priority queue and expands the one with the smallest cost lower bound. A feasible complete route gives an upper bound. Branches that cannot improve it or meet the deadline are discarded. Exact integer milli-INR costs avoid floating-point pruning errors.

Working of the Algorithm

- 1. Try two greedy tours to obtain an initial feasible upper bound.
- 2. Branch by appending an unvisited city. Bound remaining cost using outgoing and incoming edge minima.
- 3. Use the larger minimum sum, not their sum; add the incurred cost. Apply a separate time lower bound.
- 4. Check the return edge on completion. Stop only when no live node can improve the best feasible tour.

Complexity Analysis

Bounding takes O(N squared) per generated partial route. For G generated prefixes and peak queue size Q, search takes O(G * (N squared + log(Q+1))). Pruning may greatly reduce G, but its worst-case growth remains factorial.

Time Complexity

Best Case: O(N squared) when a feasible seed matches the root lower bound.

Average Case: Depends on the graph, deadline and bound strength; no fixed polynomial guarantee.

Worst Case: Factorial search, with bounding and priority-queue overhead.

Space Complexity

O(N squared + Q*N) for road matrices and queued partial paths. Hard 20+ city inputs may still require substantial time and memory.

Algorithm - Steps

- Step 1: Read roads, fuel price, starting city and total time limit.
- Step 2: Use a min-priority queue to expand the least-bound partial route.
- Step 3: Prune impossible or uncompetitive routes; print the proven optimum or infeasibility.

Pseudocode

```
initialize bestCost from any feasible greedy seed
insert the root if its bounds allow improvement
while the queue is not empty:
    remove the node with the least cost lower bound
    if its bound >= bestCost: stop
    extend to each unvisited reachable city
    test complete tours including the return edge
    queue only children that can improve and meet the deadline
```

Flow of Execution

START -> Read roads -> Seed upper bound -> Expand least-bound node -> Prune or branch -> Prove best tour -> END.

6. Example

At fuel price 100 INR/litre and a 90-minute limit, route 1 -> 3 -> 2 -> 4 -> 1 uses 5.5 litres, costs 550 INR and takes 75 minutes. A 130-minute limit permits the cheaper 1 -> 2 -> 3 -> 4 -> 1 tour: 400 INR and 120 minutes. A 70-minute limit has no feasible tour.

Conclusion

LC Branch and Bound proves an optimal fuel-cost tour under the entered time constraint when the search finishes. Safe lower bounds reduce unnecessary work, while worst-case factorial complexity remains. Road data are a static snapshot; changed conditions require a fresh run.